Course Code: MLA 0305

Subject: REINFORCEMENT LEARNING

Name: MICHELLE JOANNA G

Reg No: 192325005

Title: Digital Forensics using TD(0), SARSA, and Q-Learning

1. Pseudo Code

Algorithm 1: TD(0) for Digital Forensics

Input:

Forensic states S

Learning rate α

Discount factor γ

Initialize V(s)=0 for all states

Repeat until investigation ends

Observe current state St

Select next state St+1

Receive reward Rt+1

Update value function

$V(S_t)=V(S_t)+\alpha[R_{t+1}+\gamma V(S_{t+1})-V(S_t)]$

End Repeat

Output:

Estimated forensic state values

Algorithm 2: SARSA

Input:

State space S

Action space A

Learning rate α

Discount factor γ

Exploration ε

Initialize Q(s,a)

Choose action using ε-greedy

Repeat

Execute action At

Observe reward Rt+1

Observe next state St+1

Choose next action At+1

Update Q-table

Q(St,At)=Q(St,At)+α[

Rt+1+γQ(St+1,At+1)-Q(St,At)]

Until investigation completed

Output:

Optimal forensic policy

Algorithm 3: Q-Learning

Input:

States S

Actions A

Learning rate α

Discount factor γ

Initialize Q-table

Repeat

Observe current state

Choose action

Receive reward

Observe next state

Update

Q(St,At)=Q(St,At)+α[

Rt+1+γmaxQ(St+1,a)-Q(St,At)]

Until convergence

Output:

Optimal forensic investigation strategy

2. Algorithm with Equations

Step 1: Initialize Digital Forensic Environment

Define

$$S = \{s1, s2, s3 \dots \dots \dots, sn\} \qquad (1)$$

where

- **S** = Set of forensic investigation states
- **n** = Total number of states

Eq. (1): Defines all possible investigation states such as evidence acquisition, preprocessing, analysis, and reporting.

Step 2: Define Action Space

$$\mathrm{A} = \{\mathrm{a1}, \mathrm{a2}, \mathrm{a3} \dots \dots \dots, \mathrm{am}\} \qquad (2)$$

where

- **A** = Set of actions
- **m** = Number of forensic actions

Examples:

- Evidence Collection
- Malware Analysis
- Log Examination

- Report Generation

Eq. (2): Represents all actions available to the digital forensic investigator.

Step 3: Define Reward Function

$$R(s, a) = Accuracy - Cost \quad (3)$$

where

- **R** = Reward
- Accuracy = Investigation accuracy
- Cost = Computational cost

Eq. (3): Rewards accurate evidence identification while minimizing investigation cost.

Step 4: TD(0) Value Update

$$V(st) = V(st) + \alpha[Rt + 1 + \gamma V(st + 1) - V(st)] \quad (4)$$

where

- **V(s)** = State value
- **α** = Learning rate
- **γ** = Discount factor
- **R** = Reward

Eq. (4): Updates the value of the current forensic state using the immediate reward and estimated future value.

Step 5: SARSA Update

$$Q(st, at) = Q(st, at) + \alpha[Rt + 1 + \gamma Q(st + 1, at + 1) - Q(st, at)] \quad (5)$$

where

- **Q(s,a)** = State-action value
- **α** = Learning rate
- **γ** = Discount factor

Eq. (5): Updates the Q-value using the action actually selected in the next state, making SARSA an on-policy learning algorithm.

Step 6: Q-Learning Update

$$Q(st, at) = Q(st, at) + \alpha[Rt + 1 + \gamma amaxQ(st + 1, a) - Q(st, at)] \quad (6)$$

where

- **max Q** = Maximum future Q-value

Eq. (6): Updates the Q-value using the best possible future action, making Q-Learning an off-policy algorithm.

Step 7: Prediction

$$Policy = \arg maxQ(s, a) \quad (7)$$

where

- **Policy** = Best forensic decision
- **arg max** = Action with highest Q-value

Eq. (7): Selects the optimal action for evidence analysis based on the learned Q-values.

3. Abbreviations

Symbol	Meaning
S	State Space
A	Action Space
R	Reward
V(s)	State Value Function
Q(s,a)	Action Value Function
α	Learning Rate
γ	Discount Factor
ε	Exploration Rate
t	Time Step
Policy (π)	Decision Strategy

4. Steps for Calculation

Step 1

Initialize all state values and Q-values to zero.

Step 2

Observe the current forensic investigation state.

Step 3

Choose an action (e.g., evidence collection, malware analysis, log inspection).

Step 4

Execute the action and receive a reward based on investigation success.

Step 5

Move to the next investigation state.

Step 6

Update the values using:

- **Eq. (4)** for TD(0),
- **Eq. (5)** for SARSA,
- **Eq. (6)** for Q-Learning.

Step 7

Repeat until the investigation converges or all evidence has been analyzed.

5. Prediction Process

1. Load new digital forensic evidence.
2. Determine the current investigation state.
3. Select the best action using the learned policy (Eq. 7).
4. Estimate the expected reward.
5. Prioritize evidence based on Q-values.
6. Generate the final forensic decision (e.g., malware detected, benign file, suspicious activity).
7. Produce the investigation report with the optimal evidence sequence.